

# Solutions to Probability Exercises

1. To show  $A^c$  and  $B^c$  are independent we need to show  $P(A^c \cap B^c) = P(A^c)P(B^c)$ .

$$\begin{aligned}P(A^c \cap B^c) &= P[(A \cup B)^c] \\&\quad \text{using De Morgan's Law} \\&= 1 - P(A \cup B) \\&= 1 - [P(A) + P(B) - P(A \cap B)] \\&= 1 - P(A) - P(B) + P(A \cap B) \\&\quad \text{since } A \text{ and } B \text{ are independent} \\&= P(A^c) - P(B)[1 - P(A)] \\&= P(A^c)[1 - P(B)] \\&= P(A^c)P(B^c)\end{aligned}$$

Hence,  $A^c$  and  $B^c$  are independent.

2. There are multiple ways of approaching this, but maybe the simplest is to note that

$$\begin{aligned}P(A | B) + P(A^c | B) &= \frac{P(A \cap B)}{P(B)} + \frac{P(A^c \cap B)}{P(B)} \\&\quad \text{since we are given that } P(B) > 0 \\&= \frac{P((A \cap B) \cup (A^c \cap B))}{P(B)} \\&\quad \text{since } A \cap B \text{ and } A^c \cap B \text{ are disjoint (mutually exclusive)} \\&= \frac{P(B)}{P(B)} \\&= 1\end{aligned}$$

The desired result follows directly.

3. Define  $C, F_1, F_2$ , and  $F_3$  as the following events:

$$\begin{aligned}C &= \{\text{Member has hospital expenses exceeding \$100,000}\} \\F_1 &= \{\text{Member is under the age of 40}\} \\F_2 &= \{\text{Member is between the age of 40 and 65}\} \\F_3 &= \{\text{Member is over 65}\}\end{aligned}$$

From the given information, we know that

$$P(C | F_1) = 0.03, \quad P(C | F_2) = 0.04, \quad P(C | F_3) = 0.10, \quad \text{and} \quad P(F_3) = 0.20.$$

To avoid paying an additional reinsurance premium, the HMO requires that  $P(C) \leq 0.05$ . By a Law of Total Probability,

$$\begin{aligned} P(C) &= P(C | F_1)P(F_1) + P(C | F_2)P(F_2) + P(C | F_3)P(F_3) \\ &= (0.03)P(F_1) + (0.04)P(F_2) + (0.10)(0.20) \\ &= (0.03)P(F_1) + (0.04)P(F_2) + 0.02. \end{aligned}$$

So, the requirement  $P(C) \leq 0.05$  is equivalent to

$$(0.03)P(F_1) + (0.04)P(F_2) \leq 0.03;$$

or,

$$3P(F_1) + 4P(F_2) \leq 3.$$

To ease notation, define  $x = P(F_1)$ . Then

$$P(F_2) = 1 - P(F_1) - P(F_3) = 1 - x - 0.2 = 0.8 - x.$$

Thus, we need

$$3x + 4(0.8 - x) \leq 3,$$

which is true when  $x \geq 0.2$ .

Conclude that to avoid exceeding the 5% trigger point, at least 20% of the HMO's members should be under the age of 40.

4. Let  $C_1$ ,  $C_2$ , and  $C_3$  denote the events of drawing the loaded heads coin, the loaded tails coin, and the fair coin, respectively. Let  $H$  be the event of getting heads on the toss. So, we want to find  $P(C_3 | H)$ .

We note that the calculation of the reverse conditional  $P(H | C_3)$  is “easy,” hence we immediately think to use Bayes’ Rule:

$$\begin{aligned} P(C_3 | H) &= \frac{P(H | C_3)P(C_3)}{P(H | C_1)P(C_1) + P(H | C_2)P(C_2) + P(H | C_3)P(C_3)} \\ &= \frac{\left(\frac{1}{2}\right)\left(\frac{1}{3}\right)}{(1)\left(\frac{1}{3}\right) + (0)\left(\frac{1}{3}\right) + \left(\frac{1}{2}\right)\left(\frac{1}{3}\right)} \\ &= \frac{1}{3}. \end{aligned}$$

5. Start by using the rule for conditional probability:

$$\begin{aligned}
P(A_2 | A_1 \cap B) &= \frac{P(A_2 \cap A_1 \cap B)}{P(A_1 \cap B)} \\
&\quad \text{since } P(A_1 \cap B) > 0 \\
&= \frac{P(A_2 \cap A_1 | B)P(B)}{P(A_1 \cap B)} \\
&= \frac{P(A_2 | B)P(A_1 | B)P(B)}{P(A_1 \cap B)} \\
&\quad \text{since } A_1, A_2 \text{ independent conditional on } B \\
&= \frac{P(A_2 | B)P(A_1 \cap B)}{P(A_1 \cap B)} \\
&= P(A_2 | B)
\end{aligned}$$

The intuition here is that if  $A_1$  and  $A_2$  are independent conditional on  $B$ , then once you are given that event  $B$  has occurred, there is no additional “information” regarding the probability of  $A_2$  provided by conditioning on  $A_1$ .

6. We know that all of the probabilities must sum to one, i.e.,

$$\sum_{k=0}^{\infty} f_X(k) = 1.$$

We can also see that

$$\begin{aligned}
f_X(1) &= \frac{1}{2}f_X(0) \\
f_X(2) &= \frac{1}{2}f_X(1) = \left(\frac{1}{2}\right)^2 f_X(0)
\end{aligned}$$

and it becomes clear that, for  $k = 0, 1, 2, \dots$ ,

$$f_X(k) = \left(\frac{1}{2}\right)^k f_X(0).$$

So, now we can say that it must hold that

$$1 = \sum_{k=0}^{\infty} f_X(k) = \sum_{k=0}^{\infty} \left(\frac{1}{2}\right)^k f_X(0).$$

It follows that

$$f_X(0) = \left( \sum_{k=0}^{\infty} \left( \frac{1}{2} \right)^k \right)^{-1} = \frac{1}{2}.$$

And finally we can claim that, for  $k = 0, 1, 2, \dots$ ,

$$f_X(k) = \left( \frac{1}{2} \right)^k f_X(0) = \left( \frac{1}{2} \right)^{k+1}.$$

7. Let  $X$  equal the number of upward moves in 7 periods. Since the moves are independent, and the probability of a move up is 0.4 on each step, it follows that  $X$  has the binomial(7, 0.4) distribution.

Next note that

$$S_7 = 4(2)^X(0.5)^{7-X}$$

and that  $S_7 \geq 32$  if and only if  $X \geq 5$ .

The desired probability follows using the binomial pmf:

$$P(S_7 \geq 32) = P(X \geq 5) = \sum_{k=5}^7 \binom{7}{k} (0.4)^k (0.6)^{7-k} = 0.096.$$

8. Begin by noting that if  $X$  has the Geometric( $p$ ) distribution, then, for  $k = 0, 1, 2, \dots$ ,

$$\begin{aligned} P(X \geq k) &= P(\text{There are at least } k \text{ failures before the first success}) \\ &= P(\text{The first } k \text{ trials are failures}) \\ &= (1 - p)^k. \end{aligned}$$

Now we can use the rule for conditional probability to see the memoryless property. For any nonnegative integers  $n$  and  $m$ ,

$$\begin{aligned} P(X \geq n + m \mid X \geq n) &= \frac{P(X \geq n + m \text{ and } X \geq n)}{P(X \geq n)} \\ &\quad (\text{since } p < 1, \text{ it holds that } P(X \geq n) > 0) \\ &= \frac{P(X \geq n + m)}{P(X \geq n)} \\ &= \frac{(1 - p)^{n+m}}{(1 - p)^n} \\ &= (1 - p)^m \\ &= P(X \geq m) \end{aligned}$$

9. Let  $X$  equal the number of bacteria in a single 10 ml sample. We are given that  $X$  has the Poisson( $\lambda = 1.4$ ) distribution.

Let the random variable  $Y$  be the number of samples out of the 20 that are contaminated. We can assume that samples are independent, so  $Y$  is going to have a binomial distribution with  $n = 20$  and

$$\begin{aligned}
 p &= P(\text{individual sample is contaminated}) \\
 &= P(X \geq 2) \\
 &= 1 - P(X = 0) - P(X = 1) \\
 &= 1 - \frac{e^{-1.4}1.4^0}{0!} - \frac{e^{-1.4}1.4^1}{1!} \\
 &\quad \text{plugging in to the Poisson pmf} \\
 &= 1 - 2.4e^{-1.4}.
 \end{aligned}$$

Finally, we can just use the binomial pmf to find that

$$P(Y = 10) = \binom{20}{10} (1 - 2.4e^{-1.4})^{10} (2.4e^{-1.4})^{10} = 0.125.$$

10. (a) The “trick” in this proof is the first two steps.

$$\sum_{i=1}^{\infty} P(X \geq i) = \sum_{i=1}^{\infty} \sum_{k=i}^{\infty} P(X = k)$$

$$= \sum_{k=1}^{\infty} \sum_{i=1}^k P(X = k)$$

the order of summation can be switched because the terms are all nonnegative

$$= \sum_{k=1}^{\infty} k P(X = k)$$

because  $P(X = k)$  does not depend on  $i$

$$= \sum_{k=0}^{\infty} k P(X = k)$$

since for  $k = 0$ ,  $k P(X = k) = 0$  as well

$$= E(X)$$

- (b) In Exercise 8 above we argued that if  $X \sim \text{Geometric}(p)$  then  $P(X \geq k) = (1-p)^k$ . Now, using the result from part (a),

$$E(X) = \sum_{i=1}^{\infty} P(X \geq i) = \sum_{i=1}^{\infty} (1-p)^i = \frac{1-p}{p}$$

where this final equality follows from a result regarding geometric series.

11. We have that for  $x \geq 0$ ,

$$F_X(x) = 1 - e^{-2x} (1 + 2x + 2x^2).$$

Thus, for  $x \geq 0$ ,

$$f_X(x) = \frac{d}{dx} F_X(x) = 2e^{-2x} (1 + 2x + 2x^2) - e^{-2x} (2 + 4x) = 4x^2 e^{-2x}.$$

For  $x < 0$ ,  $F_X(x) = 0$ , so  $f_X(x) = 0$ .

This matches the density of the  $\text{Gamma}(3,2)$  distribution.

12. It helps to rewrite the pdf  $f_X(x)$  as follows:

$$f_X(x) = \begin{cases} 0.5e^x & x < 0 \\ 0.5e^{-x} & x \geq 0 \end{cases},$$

so that now we can find

$$\begin{aligned} P(2.5 \leq |X| \leq 3.5) &= P(-3.5 \leq X \leq -2.5 \text{ or } 2.5 \leq X \leq 3.5) \\ &= P(-3.5 \leq X \leq -2.5) + P(2.5 \leq X \leq 3.5) \\ &= \int_{-3.5}^{-2.5} 0.5e^x dx + \int_{2.5}^{3.5} 0.5e^{-x} dx \\ &= 0.5e^x \Big|_{-3.5}^{-2.5} - 0.5e^{-x} \Big|_{2.5}^{3.5} \\ &= 0.5(e^{-2.5} - e^{-3.5}) - 0.5(e^{-3.5} - e^{-2.5}) \\ &= e^{-2.5} - e^{-3.5} \\ &= 0.052. \end{aligned}$$

13. Let  $X$  denote the lifetime of the component. Solely to make things a little cleaner, let  $X$  be in units of 1000s of hours. (The choice of units is unimportant, provided you are consistent throughout.)

Thus, we are asked to initially assume that  $X$  has the exponential( $\lambda = 0.1$ ) distribution. (Remember, the *mean* is 10000 hours, so we must choose  $\lambda$  so that  $E(X) = 1/\lambda = 10$ .)

So, we can derive the desired probability as

$$\begin{aligned} P(X \leq 11 \mid X > 9) &= 1 - P(X > 11 \mid X > 9) \\ &\quad \text{using the result from Exercise 2 above} \\ &= 1 - P(X > 2) \\ &\quad \text{using the memoryless property of the exponential distribution} \\ &= F_X(2) \\ &= 1 - e^{-0.2} \\ &= 0.18. \end{aligned}$$

If instead we assume that  $X$  has the uniform(8, 12) distribution, then we lose the memoryless property. Thus,

$$P(X \leq 11 \mid X > 9) = \frac{P(9 < X \leq 11)}{P(X > 9)} = \frac{1/2}{3/4} = \frac{2}{3}$$

where these probabilities are found by simply using the fact that the interval (9, 11) is half of the support of the random variable, for example.

14. (a) Here we will let  $I_A(y)$  denote an indicator function, i.e.,  $I_A(y)$  evaluates to one if  $y \in A$ , and to zero otherwise.

$$\begin{aligned} \int_{t=0}^{\infty} P(Y > t) dt &= \int_{t=0}^{\infty} \left[ \int_{y=t}^{\infty} f_Y(y) dy \right] dt \\ &= \int_{t=0}^{\infty} \left[ \int_{y=0}^{\infty} I_{(t, \infty)}(y) f_Y(y) dy \right] dt \\ &= \int_{y=0}^{\infty} \left[ \int_{t=0}^{\infty} I_{(t, \infty)}(y) f_Y(y) dt \right] dy \end{aligned}$$

since the integrand is nonnegative, we can interchange the order of integration

$$\begin{aligned} &= \int_{y=0}^{\infty} \int_{t=0}^y f_Y(y) dt dy \\ &\quad \text{since } I_{(t, \infty)}(y) = 1 \text{ for } 0 < t < y \\ &= \int_{y=0}^{\infty} f_Y(y) \left[ \int_{t=0}^y dt \right] dy \\ &= \int_{y=0}^{\infty} y f_Y(y) dy \\ &= E(Y) \end{aligned}$$

since we are given that  $P(Y < 0) = 0$

- (b) If  $X \sim \text{exponential}(\lambda)$ , then  $P(X > t) = e^{-\lambda t}$  for  $t \geq 0$ . It holds that  $P(X < 0) = 0$ , so we can use part (a) to find

$$E(X) = \int_0^{\infty} e^{-\lambda t} dt = -\frac{e^{-\lambda t}}{\lambda} \Big|_0^{\infty} = \frac{1}{\lambda}.$$

15. (a)

$$\begin{aligned}
 E[g(X)] &= \int_{-\infty}^{\infty} g(x) f_X(x) dx \\
 &\quad \text{using the rule for the lazy statistician} \\
 &= \int_{-\infty}^{\infty} \min(x, m) f_X(x) dx \\
 &= \int_{-\infty}^m \min(x, m) f_X(x) dx + \int_m^{\infty} \min(x, m) f_X(x) dx \\
 &= \int_{-\infty}^m x f_X(x) dx + \int_m^{\infty} m f_X(x) dx \\
 &= \int_{-\infty}^m x f_X(x) dx + m P(X > m) \\
 &= \int_{-\infty}^m x f_X(x) dx + m[1 - F_X(m)]
 \end{aligned}$$

(b) Evaluate  $\int_0^m [1 - F_X(x)] dx$  using integration by parts. (In the standard notation used when integrating by parts, we are setting  $u = 1 - F_X(x)$  and  $v = x$ .)

$$\begin{aligned}
 \int_0^m [1 - F_X(x)] dx &= [1 - F_X(x)]x \Big|_0^m - \int_0^m (-f_X(x))x dx \\
 &= m[1 - F_X(m)] + \int_0^m x f_X(x) dx \\
 &= m[1 - F_X(m)] + \int_{-\infty}^m x f_X(x) dx \\
 &\quad \text{since we are given that } X \text{ is nonnegative} \\
 &= E[g(X)] \\
 &\quad \text{using the result from part (a)}
 \end{aligned}$$

(c) Let  $X$  be the size of the loss in thousands of dollars. We are given that  $X$  has the exponential(0.5) distribution. The insurer would like to choose the  $\text{cap}(m)$  so that  $E[\min(X, m)] = 1.5$ .

Since  $X$  is nonnegative and  $m \geq 0$ , we can use the result in part (b).

Recall that if  $X$  has the exponential( $\lambda$ ) distribution, then for  $x \geq 0$ ,

$$F_X(x) = P(X \leq x) = 1 - e^{-\lambda x}.$$

In this case  $\lambda = 0.5$ , so

$$E[\min(X, m)] = \int_0^m e^{-0.5x} dx = -2e^{-0.5x} \Big|_0^m = 2(1 - e^{-0.5m})$$

Thus, we need to find  $m$  that satisfies

$$2(1 - e^{-0.5m}) = 1.5.$$

This holds when

$$m = 2 \ln 4 = 2.773$$

and the required cap is \$2773.

16. Let  $T = \min(X, Y)$ . We will derive the CDF of  $T$  to determine its distribution. For  $t \leq 0$ ,  $P(T \leq t) = 0$ .

For  $t > 0$ ,

$$\begin{aligned} P(T \leq t) &= 1 - P(T > t) \\ &= 1 - P(X > t \text{ and } Y > t) \\ &= 1 - P(X > t)P(Y > t) \\ &\quad \text{since } X, Y \text{ independent} \\ &= 1 - e^{-\lambda_1 t} e^{-\lambda_2 t} \\ &= 1 - e^{-(\lambda_1 + \lambda_2)t} \end{aligned}$$

Hence, by the uniqueness of the CDF,  $T \sim \text{exponential}(\lambda_1 + \lambda_2)$ .

17. Clearly  $g(x) \geq 0$  for all  $x$ , since  $f(x) \geq 0$  and  $0 \leq F(x) \leq 1$ .

So, we only need to show that

$$\int_{-\infty}^{\infty} g(x) dx = \int_{-\infty}^{\infty} 2f(x)F(x) dx = 1.$$

Using the substitution  $u = F(x)$  we note that  $du = f(x) dx$  and, therefore,

$$\int_{-\infty}^{\infty} 2f(x)F(x) dx = \lim_{T \rightarrow \infty} \int_{-T}^T 2f(x)F(x) dx = \lim_{T \rightarrow \infty} \int_{F(-T)}^{F(T)} 2u du = \int_0^1 2u du = 1.$$

18. We have a heuristic notion that if a game presents us with positive expected winnings, then that game is a good bet. This exercise illustrates that this idea is not quite so simple.

Let  $Y$  denote the number of tosses required to see the first tails, so that  $X = 2^Y$ .

- (a) For  $k = 1, 2, \dots$ ,

$$P(X = 2^k) = P(Y = k) = \left(\frac{1}{2}\right)^k.$$

Thus,

$$E(X) = \sum_{k=1}^{\infty} 2^k \left(\frac{1}{2}\right)^k = \infty.$$

This implies that, no matter the cost of playing the game, my expected winnings will be positive (in fact, infinite).

- (b) Suppose that the cost of playing the game is \$1 million. The probability that I will make money on one play of the game is

$$P(X > 1000000) = P(Y \geq 20) = P(\text{first 19 tosses are heads}) = \left(\frac{1}{2}\right)^{19} \approx 2 \times 10^{-6}.$$

Thus, even though the expected winnings for the game is infinite, it seems unwise to spend \$1 million to play the game, since I am almost certain to lose money. In fact, I'm likely to lose almost all of the \$1 million.

This problem is referred to as “The St. Petersburg Paradox” and continues to be the source of research and discussion. See, for example,

<https://plato.stanford.edu/entries/paradox-stpetersburg/>.

19. Let  $X$  denote the total cost of completing the repair, including any diagnostic checks. Let  $A$  be the event that “cause 1” is the actual source of the breakdown. So, we know that  $P(A) = p$ .

If we check for “cause 1” first it holds that

$$\begin{aligned} E(X) &= E(X | A)P(A) + E(X | A^c)P(A^c) \\ &\quad \text{using a Law of Total Probability} \\ &= (C_1 + R_1)p + (C_1 + R_2)(1 - p) \\ &= p(R_1 - C_2 - R_2) + C_1 + C_2 + R_2. \end{aligned}$$

If we check for “cause 2” first it holds that

$$\begin{aligned} E(X) &= E(X | A)P(A) + E(X | A^c)P(A^c) \\ &= (C_2 + C_1 + R_1)p + (C_2 + R_2)(1 - p) \\ &= p(C_1 + R_1 - R_2) + C_1 + R_2. \end{aligned}$$

We want to determine when it’s best to check for “cause 1” prior to “cause 2.” This would be the case when

$$p(R_1 - C_2 - R_2) + C_1 + C_2 + R_2 < p(C_1 + R_1 - R_2) + C_1 + R_2.$$

Rearranging, we see that this holds when

$$p > \frac{C_1}{C_1 + C_2}.$$

It is not surprising that this result does not depend on  $R_1$  and  $R_2$  since the repair is only performed for problem that is found; it is only a diagnostic check which is potentially going to be “wasted effort.”

20. If he were randomly guessing, he has probability 0.5 of getting each question right. So  $X$ , the number of questions he gets correct under random guessing, follows the binomial distribution with parameters  $n = 10$  and  $p = 0.5$ . Thus

$$P(X = 8) = \binom{10}{8} (0.5)^8 (0.5)^2 = 45(0.5)^{10} = 0.0439.$$

Similarly, we can compute  $P(X = 9) = 0.0098$ , and  $P(X = 10) = 0.0010$ . So, the probability that the respondent would have done at least as well if he had no ESP is 0.0547.

21. (a) We first note that  $f_X(x) \geq 0$  for all  $x$ , and then that

$$\int_{-\infty}^{\infty} f_X(x) dx = \int_{\theta}^{\infty} \left( \frac{k\theta^k}{x^{k+1}} \right) dx = k\theta^k \left[ -\frac{x^{-k}}{k} \right]_{\theta}^{\infty} = 1$$

Hence  $f_X$  satisfies the properties of being a density.

- (b) For any  $x < \theta$ , we know that  $P(X \leq x) = 0$ , and for any  $x \geq \theta$ ,

$$P(X \leq x) = \int_{-\infty}^x f_X(u) du = \int_{\theta}^x \left( \frac{k\theta^k}{u^{k+1}} \right) du = k\theta^k \left[ -\frac{u^{-k}}{k} \right]_{\theta}^x = 1 - \left( \frac{\theta}{x} \right)^k.$$

The CDF is

$$F_X(x) = \begin{cases} 0, & x < \theta \\ 1 - \left( \frac{\theta}{x} \right)^k, & x \geq \theta \end{cases}$$

Please remember that it is **not** correct to write that the CDF is  $F_X(x) = 1 - \left( \frac{\theta}{x} \right)^k$ .

(c) The median  $m$  will be the value that satisfies  $F_X(m) = 0.5$ , i.e.,

$$1 - \left(\frac{\theta}{m}\right)^k = 0.5.$$

This holds when  $m = \theta 2^{1/k}$ .

22. The function  $g(x) = 1/x$  is such that  $g''(x) = 2/x^3 > 0$  for  $x > 0$ . Hence, this function is convex on the support of  $X$ . Conclude from Jensen's inequality that  $E(1/X) \geq 1/E(X)$ .

23. (a) The marginal for  $X$  is as follows:

$$\begin{aligned} f_X(x) &= \int_{-\infty}^{\infty} f_{X,Y}(x,y) dy \\ &= \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \frac{1}{\pi} dy \\ &= \begin{cases} \frac{2}{\pi} \sqrt{1-x^2}, & -1 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases} \end{aligned}$$

The marginal for  $Y$  can similarly be found to be

$$f_Y(y) = \begin{cases} \frac{2}{\pi} \sqrt{1-y^2}, & -1 \leq y \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

(b) Consider the case where  $0 < d < 1$ :

$$F_D(d) = P(\sqrt{X^2 + Y^2} \leq d)$$

$$= P(X^2 + Y^2 \leq d^2)$$

$$= \underbrace{\int \int}_{x^2+y^2 \leq d^2} \frac{1}{\pi} dx dy$$

$$= \frac{1}{\pi} \underbrace{\int \int}_{x^2+y^2 \leq d^2} dx dy$$

$$= d^2$$

since the double integral is the area of a circle of radius  $d$ .

Hence

$$F_D(d) = \begin{cases} 0, & d \leq 0 \\ d^2, & 0 < d < 1 \\ 1, & d \geq 1 \end{cases}$$

(c) We can find the density for  $D$  as

$$f_D(d) = \frac{\partial}{\partial d} F_D(d) = 2d \quad \text{for } 0 < d < 1$$

and hence see that

$$E(D) = \int_{-\infty}^{\infty} u f_D(u) du = \int_0^1 u \cdot 2u du = \left. \frac{2u^3}{3} \right|_0^1 = \frac{2}{3}.$$

**Or:** We could note that  $D$  has the Beta(2, 1) distribution, and the expected value can be calculated via its known properties.

**Or:** Since  $D$  is nonnegative, one could use Exercise 14 above and see that

$$E(D) = \int_0^{\infty} P(D > u) du = \int_0^1 (1 - u^2) du = \frac{2}{3}.$$

24. For  $t \leq 0$ , clearly  $F_T(t) = 0$ . For  $t > 0$ ,

$$\begin{aligned} F_T(t) &= P(T \leq t) \\ &= P(XY \leq t) \\ &= P(Y < t/X) \\ &= \int_{y=0}^{\infty} \int_{x=0}^{t/y} y e^{-y(x+1)} dx dy \\ &= \int_{y=0}^{\infty} e^{-y} \int_{x=0}^{t/y} y e^{-yx} dx dy \\ &= \int_{y=0}^{\infty} e^{-y} (1 - e^{-y \cdot t/y}) dy \\ &= (1 - e^{-t}) \int_{y=0}^{\infty} e^{-y} dy \\ &= 1 - e^{-t} \end{aligned}$$

We recognize that this is the CDF of the Exponential(1) distribution. Hence, by the uniqueness of the CDF,  $T$  has the Exponential(1) distribution.

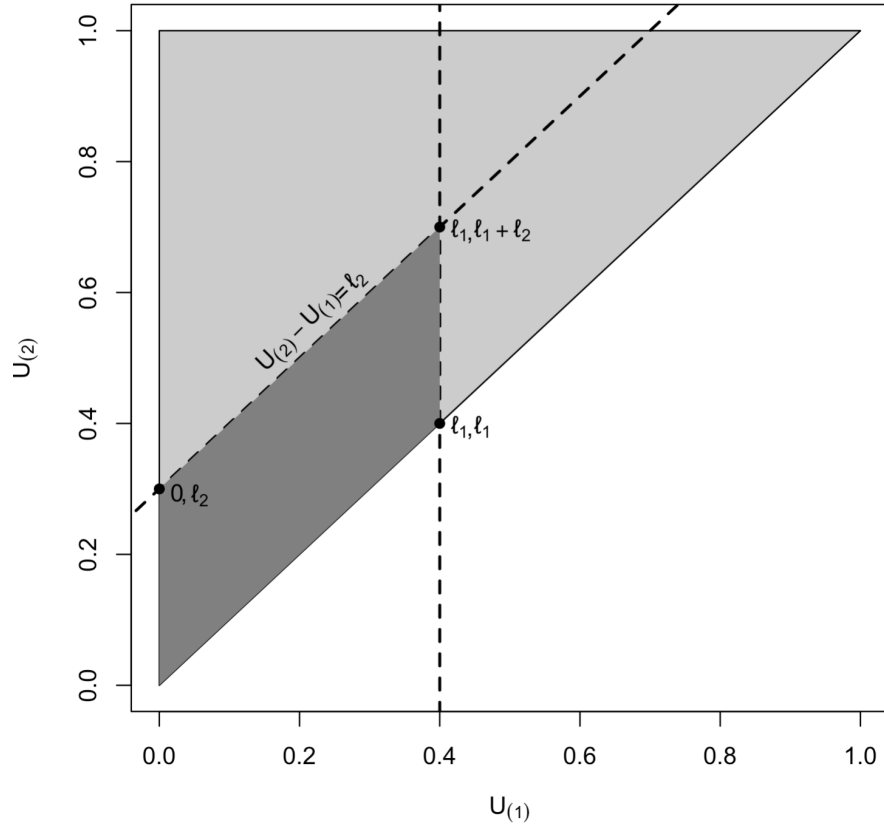
25. (a) In lecture, we established that the joint pdf of  $U_{(1)}$  and  $U_{(2)}$  is given by

$$f_{U_{(1)}, U_{(2)}}(u_1, u_2) = \begin{cases} 2 & \text{if } 0 \leq u_1 \leq u_2 \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Now, we defined  $L_1$  and  $L_2$  by  $L_1 = U_{(1)}$  and  $L_2 = U_{(2)} - U_{(1)}$ . So  $L_1 + L_2 = U_{(2)}$ . We know that the support for  $(L_1, L_2)$  will include all values of  $\ell_1$  and  $\ell_2$  such that  $0 \leq \ell_1 \leq \ell_1 + \ell_2 \leq 1$ . For these choices of  $\ell_1$  and  $\ell_2$ ,

$$P(L_1 \leq \ell_1, L_2 \leq \ell_2) = P(U_{(1)} \leq \ell_1, U_{(2)} - U_{(1)} \leq \ell_2) \quad (1)$$

Since we know that the joint density of  $U_{(1)}$  and  $U_{(2)}$  is uniform (i.e. = 2) on its support, calculating the probability in (1) will amount to computing twice the area of the region where the event in (1) holds.



The figure shows an example when  $\ell_1 = 0.4$  and  $\ell_2 = 0.3$ . The lighter shaded region is the support of  $(U_{(1)}, U_{(2)})$ , while the darker region is where the event in (1) holds. This region is a parallelogram with base  $\ell_1$  and height  $\ell_2$ ; thus, when  $0 \leq \ell_1 \leq \ell_1 + \ell_2 \leq 1$ ,

$$F_{L_1, L_2}(\ell_1, \ell_2) = P(L_1 \leq \ell_1, L_2 \leq \ell_2) = 2\ell_1\ell_2.$$

To find the joint pdf  $f_{L_1, L_2}(\ell_1, \ell_2)$ , we take  $\frac{\partial^2 F_{L_1, L_2}}{\partial \ell_1 \partial \ell_2}$ , which yields:

$$f_{L_1, L_2}(\ell_1, \ell_2) = \begin{cases} 2 & \text{if } 0 \leq \ell_1 \leq \ell_1 + \ell_2 \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

- (b) To find  $f_{L_1}(\ell_1)$  and  $f_{L_2}(\ell_2)$ , we integrate the joint distribution over the other variable in that region. For  $0 \leq \ell_1 \leq 1$ ,

$$f_{L_1}(\ell_1) = \int_0^{1-\ell_1} f_{L_1, L_2}(\ell_1, \ell_2) d\ell_2 = \int_0^{1-\ell_1} 2 d\ell_2 = 2 - 2\ell_1$$

Similarly, for  $0 \leq \ell_2 \leq 1$ ,

$$f_{L_2}(\ell_2) = \int_0^{1-\ell_2} f_{L_1, L_2}(\ell_1, \ell_2) d\ell_1 = \int_0^{1-\ell_2} 2 d\ell_1 = 2 - 2\ell_2.$$

Finally, note that the Beta(1, 2) distribution has density

$$\frac{\Gamma(3)}{\Gamma(1)\Gamma(2)} t^{1-1} (1-t)^{2-1} = 2(1-t) = 2 - 2t,$$

restricted to the interval  $0 \leq t \leq 1$ .

Hence,  $L_1$  and  $L_2$  both have a Beta(1, 2) distribution.

**Comment:** If there were  $k$  uniform cuts made, then the length of the left-most piece could still naturally be denoted  $L_1 = U_{(1)}$ , since the length of that piece is equal to the minimum of the  $k$  independently chosen Uniform(0, 1) random variables.

Hence, for  $0 < \ell < 1$ ,

$$P(L_1 \leq \ell) = 1 - P(\text{all of the } U_i > \ell) = 1 - (1 - \ell)^k,$$

and it follows that

$$f_{L_1}(\ell) = k(1 - \ell)^{k-1}.$$

For any other choice of  $\ell$ ,  $f_{L_1}(\ell) = 0$ . Hence,  $L_1$  has the Beta(1,  $k$ ) distribution.

It is not too difficult to convince yourself that, regardless of the number of cuts, the lengths of the resulting pieces are random variables with the same distribution. Hence, all pieces (marginally) have the Beta(1,  $k$ ) distribution. (If you are not convinced that this is true, imagine instead a circle of circumference one, and place  $k + 1$  uniformly chosen cuts along its perimeter. If you take the resulting pieces and “straighten them out,” the result is identical to that of breaking the stick into  $k$  pieces. Thinking of it this way makes it more clear that there is no special “starting point.”)

Unfortunately, this simple derivation does not lead to insight into the distribution of the length of the longest of the resulting  $k$  pieces. That is much more difficult.

26. (a)

$$\begin{aligned}
P(\underbrace{Y < tX}_A) &= \underbrace{\int \int}_A f_{X,Y}(x,y) dx dy \\
&= \int_{y=0}^{\infty} \int_{x=y/t}^{\infty} \lambda_1 e^{-\lambda_1 x} \cdot \lambda_2 e^{-\lambda_2 y} dx dy \\
&= \int_{y=0}^{\infty} \lambda_2 e^{-\lambda_2 y} \int_{x=y/t}^{\infty} \lambda_1 e^{-\lambda_1 x} dx dy \\
&= \lambda_2 \int_{y=0}^{\infty} e^{-\lambda_2 y} \cdot e^{-\lambda_1 y/t} dy \\
&= \lambda_2 \int_{y=0}^{\infty} e^{-(\lambda_1/t + \lambda_2)y} dy \\
&= \lambda_2 \left( \frac{1}{\lambda_1/t + \lambda_2} \right) \\
&\quad \text{since } \int_0^{\infty} e^{-ax} dx = \frac{1}{a} \text{ for any } a \\
&= \frac{t\lambda_2}{\lambda_1 + t\lambda_2}
\end{aligned}$$

(b) For  $t > 0$ ,

$$\begin{aligned}
F_T(t) &= P(T \leq t) \\
&= P\left(\frac{Y}{X} \leq t\right) \\
&= P(Y \leq tX) \\
&= \frac{t}{1+t} \\
&\quad \text{using part (a) with } \lambda_1 = \lambda_2 = 1.
\end{aligned}$$

This means that for  $t > 0$ ,

$$f_T(t) = \frac{\partial}{\partial t} F_T(t) = \frac{\partial}{\partial t} \left( \frac{t}{1+t} \right) = \frac{1}{(1+t)^2}.$$

(c) We note that for  $t$  large,

$$tf_T(t) = \frac{t}{(1+t)^2} \approx \frac{1}{t}.$$

Thus, the derivation of  $E(T)$  will involve integrating a tail that behaves like  $1/t$  as  $t$  grows. This integral does not converge, so  $E(T) = \infty$ .

**Or:** A clever way of seeing this result is to note that  $1 + T$  has the Pareto(1, 1) distribution, so  $E(1 + T) = \infty$  which implies that  $E(T) = \infty$ . (To show that  $U = 1 + T$  is Pareto, derive the CDF of  $U$ .)

27. To find the marginal for  $X$ , we first note that the support for  $X$  is  $(0, 1)$ . So, for  $0 < x < 1$ ,

$$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x, y) dy = \int_x^{x+1} 2x dy = 2x,$$

and  $f_X(x) = 0$  for all other  $x$ . Thus,  $X$  has the Beta(2, 1) distribution.

The marginal for  $Y$  has a support of  $(0, 2)$ , but the nature of the support for  $(X, Y)$  is such that for  $0 < y < 1$ ,  $x$  ranges from 0 to  $y$ , and for  $1 \leq y < 2$ ,  $x$  ranges from  $y - 1$  to 1. Thus, for  $0 < y < 1$ ,

$$f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x, y) dx = \int_0^y 2x dx = y^2$$

and for  $1 \leq y < 2$ ,

$$f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x, y) dx = \int_{y-1}^1 2x dx = 1 - (y - 1)^2 = y(2 - y).$$

Putting it together,

$$f_Y(y) = \begin{cases} y^2, & 0 < y < 1 \\ y(2 - y), & 1 \leq y < 2 \\ 0, & \text{otherwise} \end{cases}.$$

28. (a) This can be found using a Law of Total Probability, conditioning on the value of  $Y$ . For  $x > 0$ ,

$$\begin{aligned} f_X(x) &= \sum_y f_{X|Y}(x | Y = y) f_Y(y) \\ &= f_{X|Y}(x | Y = 0) f_Y(0) + f_{X|Y}(x | Y = 1) f_Y(1) \\ &= 2e^{-2x}(0.7) + \frac{1}{3}e^{-x/3}(0.3) \\ &= 1.4e^{-2x} + 0.1e^{-x/3}. \end{aligned}$$

- (b) You could again appeal to a Law of Total Probability again, or simply use the result from part (a), as follows:

$$\begin{aligned}
 P(\text{claim exceeds \$1,000}) &= P(X > 1) \\
 &= \int_1^{\infty} f_X(u) du \\
 &= \int_1^{\infty} [1.4e^{-2u} + 0.1e^{-u/3}] du \\
 &= 0.7 e^{-2u} \Big|_1^{\infty} + 0.3 e^{-u/3} \Big|_1^{\infty} \\
 &= 1 - e^{-2} - e^{1/3}
 \end{aligned}$$

29. The point of this simple exercise is that random variables can be uncorrelated but still be strongly dependent.

- (a) When  $U = -1$ , it must hold that  $V = 0$  since  $X$  and  $Y$  must have opposite signs. When  $U = 1$ , it is either the case that  $X = Y = -1$  or  $X = Y = 1$ . The joint distribution of  $U$  and  $V$  is given by the table below.

|   |    | V    |     |      |
|---|----|------|-----|------|
|   |    | -2   | 0   | 2    |
| U | -1 | 0    | 0.5 | 0    |
|   | 1  | 0.25 | 0   | 0.25 |

- (b) We can see that  $E(U) = E(V) = 0$ , and that

$$\text{Cov}(U, V) = E(UV) = (0)(0.5) + (-2)(0.25) + (2)(0.25) = 0.$$

Hence,  $U$  and  $V$  are uncorrelated.

- (c)  $U$  and  $V$  are definitely dependent. A simple way to see this is to note that

$$P(U = -1, V = -2) = 0 \quad \text{while} \quad P(U = -1)P(V = -2) = (0.5)(0.25).$$

More heuristically, if you condition on the value of  $V$ , you would know the value of  $U$ . That is an extreme form of dependence.

30. (a) The support of  $X$  is  $(0, 1)$ , so for  $0 < x < 1$ ,

$$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x, y) dy = \int_0^{1-x} 60x^2y dy = 30x^2y^2 \Big|_0^{1-x} = 30x^2(1-x)^2.$$

For any other value of  $x$ ,  $f_X(x) = 0$ . Hence  $X$  has the Beta(3, 3) distribution. The support of  $Y$  is also  $(0, 1)$ , so for  $0 < y < 1$ ,

$$f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x, y) dx = \int_0^{1-y} 60x^2y dx = 20x^3y \Big|_0^{1-y} = 20y(1-y)^3.$$

For any other value of  $y$ ,  $f_Y(y) = 0$ . Hence  $Y$  has the Beta(2, 4) distribution.

(b) We can see that  $E(X) = 1/2$ ,  $E(Y) = 1/3$ ,  $V(X) = 1/28$ , and  $V(Y) = 2/63$  using properties of the Beta distribution.

(c) We first derive

$$\begin{aligned} E(XY) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f_{XY}(x, y) dx dy \\ &= \int_0^1 \int_0^{1-y} 60x^3y^2 dx dy \\ &= \int_0^1 15x^4y^2 \Big|_0^{1-y} dy \\ &= \int_0^1 15(1-y)^4y^2 dy \\ &= 15 \int_0^1 (1-y)^4y^2 dy \\ &= 15 \left( \frac{\Gamma(3)\Gamma(5)}{\Gamma(8)} \right) \\ &\quad \text{see Math Facts} \\ &= \frac{1}{7} \end{aligned}$$

So now we can find

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y) = \frac{1}{7} - \frac{1}{6} = -\frac{1}{42}$$

and

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{V(X)V(Y)}} = -\frac{1/42}{\sqrt{1/28 \times 2/63}} = -\frac{\sqrt{1764/2}}{42} = -\frac{1}{\sqrt{2}}.$$

(d)

$$V(X + Y) = V(X) + V(Y) + 2 \text{Cov}(X, Y) = \frac{1}{28} + \frac{2}{63} - \frac{1}{21} = \frac{5}{252}.$$

31. We have that, given  $P = p$ ,  $X$  has the binomial(100,  $p$ ) distribution. We also know that  $P$  has the Beta(1, 9) distribution.

We can use iterated expectation to see that

$$E(X) = E(E(X | P)) = E(100P) = 100E(P) = 10.$$

To find  $V(X)$  we use the result

$$V(X) = E[V(X | P)] + V[E(X | P)]$$

Let's look at each of the two parts separately.

$$\begin{aligned} E[V(X | P)] &= E[100P(1 - P)] \\ &= \int_0^1 100p(1 - p) \cdot 9(1 - p)^8 dp \\ &= 900 \int_0^1 p(1 - p)^9 dp \\ &= 900 \left( \frac{\Gamma(2)\Gamma(10)}{\Gamma(12)} \right) \\ &= \frac{90}{11} \end{aligned}$$

Alternatively,

$$\begin{aligned} E[V(X | P)] &= E[100P(1 - P)] \\ &= 100E(P - P^2) \\ &= 100[E(P) - E(P^2)] \\ &= 100(E(P) - [V(P) + E(P)^2]) \\ &= 100 \left( \frac{1}{10} - \left( \frac{9}{1100} + \left( \frac{1}{10} \right)^2 \right) \right) \\ &\quad \text{using properties of the Beta distribution} \\ &= \frac{90}{11} \end{aligned}$$

Next we see that

$$\begin{aligned}
V[E(X|P)] &= V(100P) \\
&= 100^2 V(P) \\
&= 100^2 \left( \frac{9}{(1+9)^2(1+9+1)} \right) \\
&= \frac{900}{11}
\end{aligned}$$

Putting it together we get

$$V(X) = \frac{90}{11} + \frac{900}{11} = 90.$$

32. Let's start by finding the marginal for  $(X_1, X_2)$ . We know that the support is  $x_1 > 0$  and  $x_2 > 0$ , so for any such pair  $(x_1, x_2)$ ,

$$\begin{aligned}
f_{X_1, X_2}(x_1, x_2) &= \int_0^\infty 6e^{-(x_1+2x_2+3x_3)} dx_3 \\
&= 2e^{-(x_1+2x_2)} \int_0^\infty 3e^{-3x_3} dx_3 \\
&= 2e^{-(x_1+2x_2)}.
\end{aligned}$$

Then we can find the marginals for  $X_1$  and  $X_2$  as

$$\begin{aligned}
f_{X_1}(x_1) &= \int_0^\infty 2e^{-(x_1+2x_2)} dx_2 = e^{-x_1}, \quad x_1 > 0 \\
f_{X_2}(x_2) &= \int_0^\infty 2e^{-(x_1+2x_2)} dx_1 = 2e^{-2x_2}, \quad x_2 > 0.
\end{aligned}$$

You can likewise show that

$$f_{X_3}(x_3) = 3e^{-3x_3}, \quad x_3 > 0.$$

We can now see that

$$f_{X_1, X_2, X_3}(x_1, x_2, x_3) = f_{X_1}(x_1)f_{X_2}(x_2)f_{X_3}(x_3) \quad (\star)$$

for all  $x_1 > 0$ ,  $x_2 > 0$ , and  $x_3 > 0$ . If any of the  $x_i$  are less than or equal to zero, then  $(\star)$  is trivially true since both sides equal zero. Hence,  $(\star)$  holds for all  $x_1$ ,  $x_2$  and  $x_3$ , and  $X_1$ ,  $X_2$ , and  $X_3$  are independent.

33.

$$\begin{aligned}
M_X(t) &= E(e^{tX}) \\
&= E[\underbrace{E(e^{tX}|Y)}_{M_{X|Y}(t)}] \\
&= E\left[e^{tY+t^2\tau^2/2}\right] \\
&= E\left[e^{tY}e^{t^2\tau^2/2}\right] \\
&= e^{t^2\tau^2/2} \cdot M_Y(t) \\
&= e^{t^2\tau^2/2} \cdot e^{t\mu+t^2\sigma^2/2} \\
&= \exp(t\mu + t^2(\tau^2 + \sigma^2)/2)
\end{aligned}$$

This matches the MGF of the Normal( $\mu, \tau^2 + \sigma^2$ ) distribution, hence that is the distribution of  $X$ .

34. (a) For any  $0 < x_1 < 1$  and  $0 < x_2 < 1$ ,

$$\begin{aligned}
f_{X_1, X_2}(x_1, x_2) &= \int_{-\infty}^{\infty} f_{X_1, X_2, X_3}(x_1, x_2, x_3) dx_3 \\
&= \int_0^1 \frac{1}{3}(x_1 + 2x_2 + 3x_3) dx_3 \\
&= \frac{1}{3} \left( x_1 x_3 + 2x_2 x_3 + \frac{3x_3^2}{2} \Big|_0^1 \right) \\
&= \frac{1}{3} \left( x_1 + 2x_2 + \frac{3}{2} \right).
\end{aligned}$$

(b) For  $0 < x_1 < 1$ ,  $0 < x_2 < 1$ , and  $0 < x_3 < 1$ ,

$$\begin{aligned}
f_{X_3 | X_1, X_2}(x_3 | x_1, x_2) &= \frac{f_{X_1, X_2, X_3}(x_1, x_2, x_3)}{f_{X_1, X_2}(x_1, x_2)} \\
&= \frac{\frac{1}{3}(x_1 + 2x_2 + 3x_3)}{\frac{1}{3}(x_1 + 2x_2 + \frac{3}{2})} \\
&= \frac{x_1 + 2x_2 + 3x_3}{x_1 + 2x_2 + \frac{3}{2}}.
\end{aligned}$$

Thus if we condition on  $X_2 = 1/4$  and  $X_3 = 3/4$  we have

$$\begin{aligned} f_{X_3|X_1, X_2}\left(x_3 \left| \frac{1}{4}, \frac{3}{4} \right.\right) &= \frac{\frac{1}{4} + 2 \cdot \frac{3}{4} + 3x_3}{\frac{1}{4} + 2 \cdot \frac{3}{4} + \frac{3}{2}} \\ &= \frac{\frac{7}{4} + 3x_3}{\frac{13}{4}} \\ &= \frac{7 + 12x_3}{13}, \quad 0 < x_3 < 1. \end{aligned}$$

Then

$$P\left(X_3 < \frac{1}{2} \left| X_1 = \frac{1}{4}, X_2 = \frac{3}{4} \right.\right) = \int_0^{\frac{1}{2}} \left(\frac{7 + 12x_3}{13}\right) dx_3 = \frac{1}{13} \left(7x_3 + 6x_3^2 \Big|_0^{\frac{1}{2}}\right) \approx 0.385.$$

35. Note that for any random variables  $U$  and  $V$ ,  $E(U|V)$  can be formally expressed as  $E(U|\sigma(V))$ , where  $\sigma(V)$  is the  $\sigma$ -algebra generated by  $V$ . Thus, the expression  $E(E(Y|X, Z)|X)$  can be formally expressed as  $E(E(Y|\sigma(X, Z))|\sigma(X))$ .

Note also that  $\sigma(X) \subseteq \sigma(X, Z)$ . In light of a property of conditional expectation, this implies that  $E(E(Y|\sigma(X, Z))|\sigma(X)) = E(Y|X)$ .

36. For a random variable  $Y$  and a  $\sigma$ -algebra  $\mathcal{F}$ , we know that

$$E(Y) = E(E(Y|\mathcal{F})).$$

Consider the random variable  $XE(X|\mathcal{F})$ . By the above,

$$E(XE(X|\mathcal{F})) = E(E(XE(X|\mathcal{F})|\mathcal{F})).$$

We know that  $E(X|\mathcal{F})$  is always  $\mathcal{F}$ -measurable. This, in light of a property of conditional expectations given in lecture, this implies that for the inner expectation above,

$$E(XE(X|\mathcal{F})|\mathcal{F}) = E(X|\mathcal{F})E(X|\mathcal{F}) = E(X|\mathcal{F})^2$$

.

Hence,

$$E(XE(X|\mathcal{F})) = E(E(X|\mathcal{F})^2),$$

which is the desired result.

37. We use the fact that  $X$  is  $N(\mu, \sigma^2)$  to see that

$$E(Y^k) = E(e^{Xk}) = M_X(k) = e^{t\mu + t^2\sigma^2}$$

where  $M_X$  denotes the MGF of  $X$ .

38. We need to calculate  $E(XY)$ ,  $E(X)$ , and  $E(Y)$ . First,

$$\begin{aligned}
 E(XY) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f_{XY}(x, y) \, dx \, dy \\
 &= \int_0^{\infty} \int_0^y xy \left( \frac{2e^{-2y}}{y} \right) \, dx \, dy \\
 &= \int_0^{\infty} 2e^{-2y} \frac{y^2}{2} \, dy \\
 &= \int_0^{\infty} e^{-2y} y^2 \, dy \\
 &= \frac{\Gamma(3)}{2^3} \\
 &\quad \text{using the integral table on Math Facts} \\
 &= \frac{1}{4}
 \end{aligned}$$

Next we note that for  $y > 0$ ,

$$f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x, y) \, dx = \int_0^y \left( \frac{2e^{-2y}}{y} \right) \, dx = 2e^{-2y}$$

and thus  $Y$  has the Exponential(2) distribution and  $E(Y) = 0.5$ .

Finally,

$$\begin{aligned}
 E(X) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f_{XY}(x, y) \, dx \, dy \\
 &= \int_0^{\infty} \int_0^y x \left( \frac{2e^{-2y}}{y} \right) \, dx \, dy \\
 &= \int_0^{\infty} \left( \frac{2e^{-2y}}{y} \cdot \frac{y^2}{2} \right) \, dy \\
 &= \int_0^{\infty} ye^{-2y} \, dy \\
 &= \frac{\Gamma(2)}{2^2} \\
 &\quad \text{using the integral table on Math Facts} \\
 &= \frac{1}{4}
 \end{aligned}$$

Combining the three bullets gives

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y) = \frac{1}{4} - \frac{1}{4} \cdot \frac{1}{2} = \frac{1}{8}.$$

39. Since  $X_i$  are iid, and the sample size is large, we know that by the CLT that  $\bar{X}$  is approximately  $N\left(\mu, \frac{\sigma^2}{n}\right)$ . Letting  $g(x) = x^3$ , we have  $g'(x) = 3x^2$ . Thus by the Delta method, we have that

$$g(\bar{X}) = \bar{X}^3 \approx N\left(\mu^3, \frac{9\mu^4\sigma^2}{n}\right).$$

40. Let  $X_i$   $i = 1, 2, \dots, 10000$ , denote the claim sizes. The total claims exceed \$2.5 million if  $\bar{X}$  exceeds \$250. If we **assume** that the claims are independent, the CLT tells us that  $\bar{X}$  will be (approximately)  $N(240, 800^2/10000)$ , so

$$P(\text{Claims Exceed \$2.5 million}) = P(\bar{X} > 250) = P(Z > 1.25) \approx 0.89,$$

where  $Z$  denotes a standard normal random variable.

41. Suppose that  $n$  components are in stock, and let  $X_1, X_2, \dots, X_n$  be their lifetimes. The probability that they will last at least 4000 hours is

$$P\left(\sum_{i=1}^n X_i \geq 4000\right) = P(\bar{X} \geq 4000/n).$$

If we **assume** that the component lifetimes are independent, then we know that  $\bar{X}$  is approximately  $N(100, 30^2/n)$  by the CLT. Hence,

$$P(\bar{X} \geq 4000/n) \approx P\left(Z \geq \frac{4000/n - 100}{30/\sqrt{n}}\right)$$

where  $Z$  has the standard normal distribution.

In order for

$$P\left(Z \geq \frac{4000/n - 100}{30/\sqrt{n}}\right) \geq 0.95$$

we need that

$$\frac{4000/n - 100}{30/\sqrt{n}} \leq z_{0.95} = -1.65.$$

This holds when  $n \geq 44$ .

42. (a)

$$(X_2, X_3) \sim N_2 \left( \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right)$$

$X_2$  and  $X_3$  are indeed independent; when two random variables are jointly normally distributed and uncorrelated, they are independent.

(b)

$$\begin{aligned} E(X_2^2 \cdot e^{X_3}) &= E(X_2^2)E(e^{X_3}) \\ &\quad \text{since } X_2, X_3 \text{ independent} \\ &= (V(X_2) + E(X_2)^2) M_{X_3}(1) \\ &= (1 + 2^2)e^{1 \cdot 0 + 1^2 1^2 / 2} \\ &= 5e^{1/2} \end{aligned}$$

(c) We know that  $\mathbf{Y}$  is bivariate normal with mean

$$\mathbf{A}\boldsymbol{\mu} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

and variance

$$\mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^T = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0.5 \\ 0 & 1 & 0 \\ 0.5 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 0 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 4.5 \\ 4.5 & 7 \end{bmatrix}.$$

Since  $\text{Cov}(Y_1, Y_2) = 4.5 \neq 0$ , it is definitely the case that  $Y_1$  and  $Y_2$  are dependent.